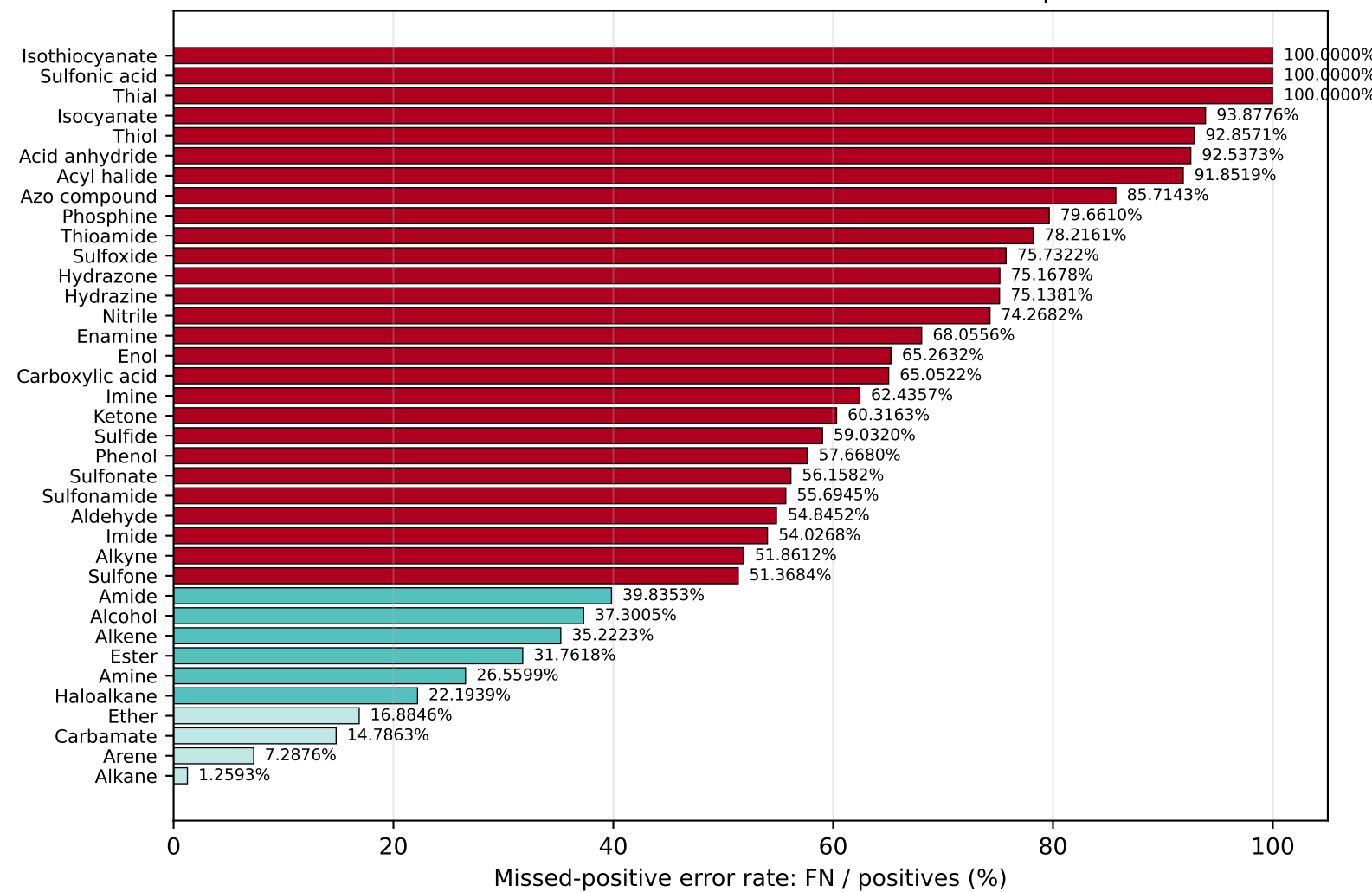
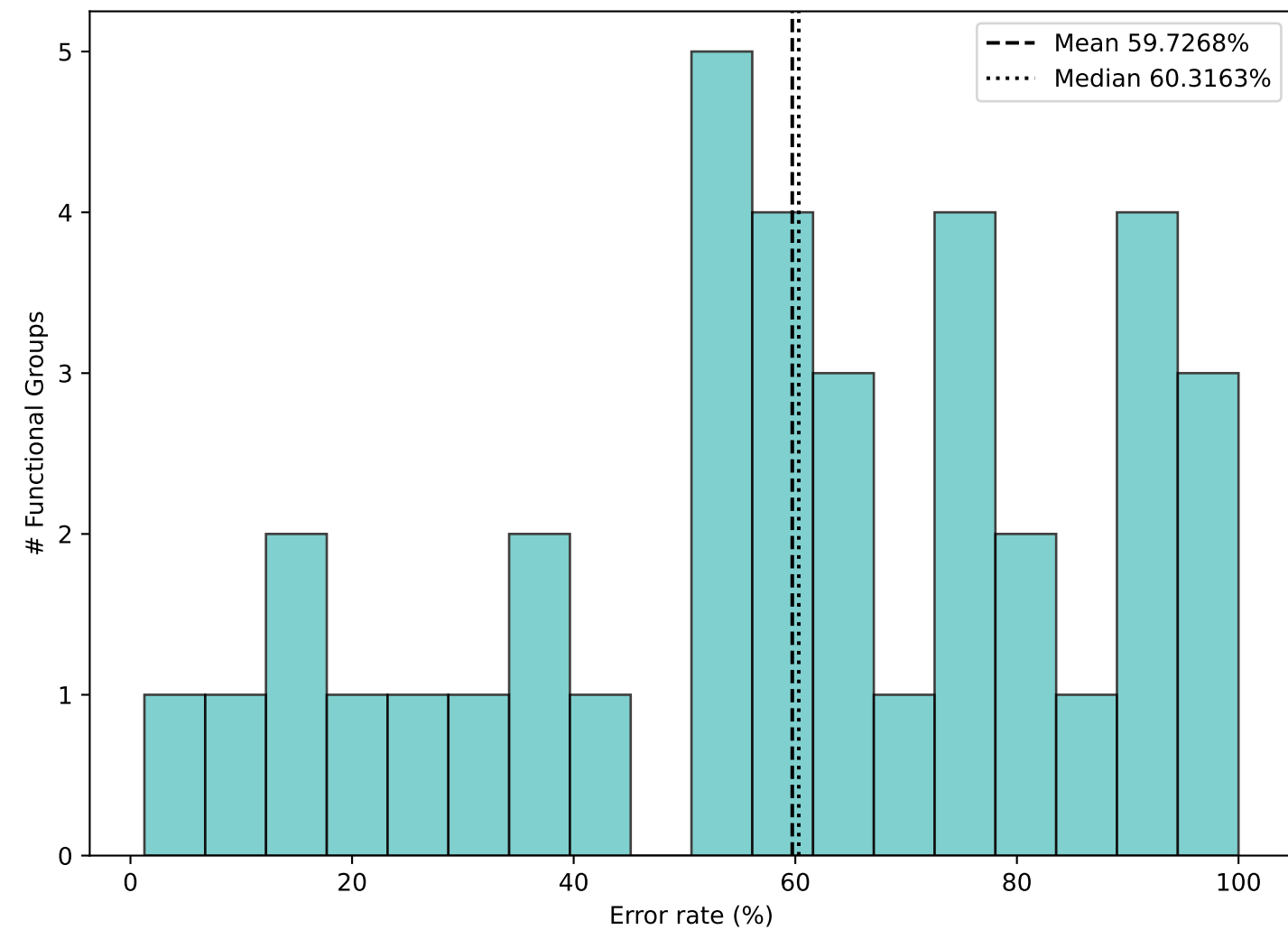


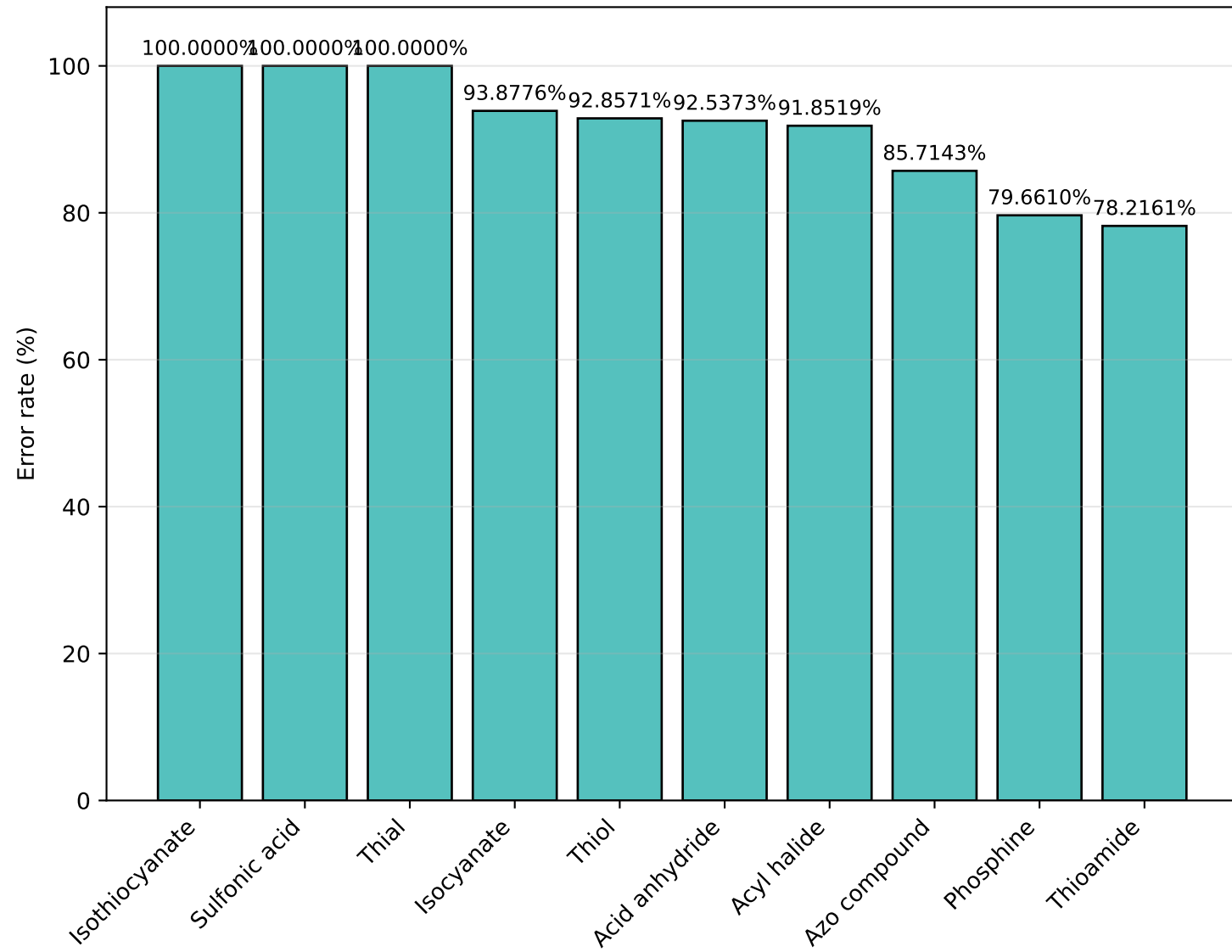
CNN H-NMR Test Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



CNN H-NMR TEST SET ERROR ANALYSIS

Global metrics:

F1 micro: 0.3844
F1 macro: 0.0435
Precision micro: 0.8890
Recall micro: 0.7681
Hamming accuracy: 0.9577

Error-rate statistics:

Mean: 59.7268%
Median: 60.3163%
Std: 27.2680%
Max: 100.0000% (Isothiocyanate)
Min: 1.2593% (Alkane)

Problem classes:

Severe (>50%): 27 groups
High (20-50%): 6 groups
Moderate (5-20%): 3 groups
Low (<=5%): 1 groups